



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 2.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MPBA G514
Course Title : Deep Learning for Business
Instructor-in-Charge : Prof. Abhishek Verma
Instructor(s) : Prof. Abhishek Verma
Tutorial/Practical Instructors: Prof. Abhishek Verma

- 1. Course description:** This course offers an introduction to deep learning and its practical relevance in business contexts. It covers foundational concepts such as neural networks, linear and logistic regression using neural architectures, deep neural networks, hyperparameter tuning, and advanced models like CNNs, RNNs, LSTMs, GANs, and deep reinforcement learning. The course blends theoretical understanding with hands-on applications to solve real-world business problems. It also highlights the challenges and limitations of deep learning methods in practice.
- 2. Scope and Objective of the Course:** The course aims to equip students with foundational and practical knowledge of deep learning algorithms and their applications in solving business challenges. The course emphasizes:
 - a. Understanding core deep learning architectures and techniques
 - b. Translating business problems into machine learning tasks
 - c. Developing, training, and tuning deep learning models using appropriate tools
 - d. Applying deep learning to solve business challenges
 - e. Utilizing TensorFlow and Keras libraries on Google Colab to implement and explore deep learning algorithms
 - f. Recognizing the limitations, ethical concerns, and interpretability issues of deep models in business decision-making
 - g. Engaging in a group-based project, where students apply deep learning techniques to some real-world business problems
- 3. Course Learning Outcomes:** By the end of this course, students will be able to:
 - a. Understand the fundamental principles of neural networks and deep learning algorithms, including their relevance to solving business problems.
 - b. Develop and train deep neural networks, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Generative Adversarial Networks (GANs), using TensorFlow and Keras on Google Colab.
 - c. Implement and fine-tune models through effective hyperparameter optimization strategies.
 - d. Use deep learning techniques to address business use cases.
 - e. Conduct group projects that involve problem formulation, data preparation, model development, and presentation of results for real-world business problems.
 - f. Assess the challenges, limitations, and ethical implications of deploying deep learning models in business problems.
 - g. Communicate technical outcomes and business insights from deep learning models to both technical and non-technical stakeholders.



4. Text Books:

1. Neural Networks and Deep Learning A Textbook by Charu Aggrawal
<https://dl.uswr.ac.ir/bitstream/Hannan/141305/2/9783319944623.pdf>
2. Chollet, F. (2021). Deep learning with Python. Simon and Schuster.

5. Reference Books:

1. Dive to Deep Learning. Available for free for learning at <https://d2l.ai/>
2. Deep Learning. Aaron Courville, Ian Goodfellow, and Yoshua Bengio available in
<https://www.deeplearningbook.org/>
3. Tensorflow, Keras & PyTorch APIs documentation

6. Course Plan:

Session No.	Lecture Session	Learning outcomes
1-6	Introduction to deep learning, and example applications	Introduction to Neural network and Deep neural network, basic functionality and applications.
7-12	Fundamentals of Neural Networks	To understand the steps involved in artificial neural networks development, Regression models.
13-14	Deep neural networks concepts and applications	To understand the data pipelines and data preprocessing steps, functions and packages in Tensorflow. To apply and transform the data using preprocessing tools in Tensorflow
15-16	Model Building & Hyper-parameters tuning	To implement a numerical prediction (regression) algorithms.
17-22	Convolution Neural Networks	To implement convolution neural networks using Keras to solve computer vision related problems such as classification and objects count.
23-30	Processing Sequences using RNNs and LSTM	To implement recurrent neural networks (RNN) and LSTM using Keras to solve sequences related problems such as time series and sentiment analysis.
31-36	Encoders and Decoders, and Transformers	To understand and implement auto-encoders, decoders and attention based transformers using Keras to solve text data problems.
37-40	Generative Adversarial Networks	To understand & apply generative adversarial networks (GANs) using Keras



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

7. **Evaluation Scheme:** Evaluation components assess a range of learning outcomes: basic comprehension of the topics taught, application of different assignments in the group projects, and ability to solve problems using deep learning. The total marks for the course are 200. The proposed evaluation scheme is given below:

Component	Duration	Weightage (%)	Date & Time	Nature of component
Mid-Sem Exam	1.5 hours	20	8-10-2026 4:00 AM to 5:30 PM	Close Book
Assignments		5	TBA in classes	Take Home
Group Project & Presentation		25	TBA in classes	Open Book
Quizzes		10	TBA in classes	Open Book
Comprehensive Exam	3 hours	40	12-12-2026 2:00 PM to 5:00 PM	Close Book

8. **Chamber Consultation Hour:** 3:30 PM to 5:00 PM, Every Wednesday

9. **Make-up Policy:** No extensions shall be granted for projects. Genuine cases such as hospitalization (upon production of the relevant documents/proof) will be considered upon discussion with individuals/team for any make ups for the final Exam.

Instructor-in-charge
Course No. MPBA G514